GTIIT Spring 2025 Calculus 1M
Workshop 10

May 12, 2025

Try to solve the following exercises. Feel free to chat about them with your classmates, but make sure to write your own solutions. If you get stuck, just move on to the next one and come back later.

If you need any help at all —whether it's a tiny hint or a big clue— don't hesitate to ask me!

Important note: If you find any errors or typos, please let me know so I can correct them.

Exercise 1. Compute the following integrals:

(i) $\int x \sin^2(x^2) dx$

(ii) $\int \sin^5(x) \cos^3(x) dx$

(iii) $\int \sin(3x) \sin(6x) dx$

Exercise 2. Prove the formula where $m, n \in \mathbb{Z}$:

$$\int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{if } m = n \end{cases}$$

Exercise 3. Use trigonometric substitution to show that

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln\left(x + \sqrt{x^2 + a^2}\right) + C$$

Exercise 4. Evaluate the integrals using the appropriate partial fraction decomposition:

(i) $\int \frac{x^2 - x + 6}{x^3 + 3x}$

(ii) $\int \frac{x^5 + x - 1}{x^3 + 1}$

(iii) $\int \frac{1}{\sqrt{x} - \sqrt[3]{x}}$, *hint: use the substitution* $u = \sqrt[6]{x}$